

Lecture 2: Introduction to Data - Part 1

LSE ME314: Introduction to Data Science and Machine Learning (<https://github.com/me314-lse>)

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What is Data?

Data vs. Information

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Data is a representation of underlying information.

Raw data may or may not contain interpretable information.

Invariably, we need to *do things to and with data* to extract signal and information.

Different Types of Data

Numerical vs. Quantitative vs. Qualitative

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Qualitative: Capturing a *quality* as a categorical variable

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But this does not mean that everything is **quantitative**!

Quantitative: Capturing a *quantity* as a continuous or discrete variable

Qualitative: Capturing a *quality* as a categorical variable

For our purposes, almost all data will need to be converted into **numerical** data

Peaches



Source: The Today Show, [Peach Benefits](#)

Quantitative Data as Numerical Data: Example

```
# Quantitative data about peach varieties:
peach_counts <- data.frame(
  variety = c("Donut", "Redhaven", "Nectarine", "White"),
  count = c(150, 200, 100, 120),
  gbp_per_lb = c(2, 3, 3.5, 2.5)
)

# Easy to calculate the total revenue from each variety:
peach_counts %>%
  mutate(revenue = count * gbp_per_lb) %>%
  select(variety, revenue) %>%
  arrange(desc(revenue))
```

	variety	revenue
1	Redhaven	600
2	Nectarine	350
3	Donut	300
4	White	300

```
# Save our data as a .csv:
write.csv(peach_counts, "../data/peach_counts.csv", row.names = FALSE)
```

Qualitative Data as Numerical Data: Example

```
# Qualitative data about peach varieties:
peach_features <- data.frame(
  variety = c("Donut", "Redhaven", "Nectarine", "White"),
  color = c("Yellow", "Red", "Red", "White"),
  size = c("Small", "Large", "Medium", "Large"),
  taste = c("Tangy", "Sweet", "Sweet", "Sweet"),
  fuzziness = c("Fuzzy", "Fuzzy", "Smooth", "Fuzzy")
)
```

peach_features

	variety	color	size	taste	fuzziness
1	Donut	Yellow	Small	Tangy	Fuzzy
2	Redhaven	Red	Large	Sweet	Fuzzy
3	Nectarine	Red	Medium	Sweet	Smooth
4	White	White	Large	Sweet	Fuzzy

Qualitative Data as Numerical Data: Example

```
# Convert the categorical features into numerical data:
# Column 1 is the variety, then a series of dummies for features:
peach_features %>%
  mutate(
    color_Yellow = ifelse(color == "Yellow", 1, 0),
    color_Red = ifelse(color == "Red", 1, 0),
    color_White = ifelse(color == "White", 1, 0),
    size_Medium = ifelse(size == "Medium", 1, 0),
    size_Large = ifelse(size == "Large", 1, 0),
    size_Small = ifelse(size == "Small", 1, 0),
    taste_Sweet = ifelse(taste == "Sweet", 1, 0),
    taste_Tangy = ifelse(taste == "Tangy", 1, 0),
    fuzziness_Fuzzy = ifelse(fuzziness == "Fuzzy", 1, 0),
    fuzziness_Smooth = ifelse(fuzziness == "Smooth", 1, 0)
  ) %>%
  select(-color, -size, -taste, -fuzziness) # remove original categorical columns
```

	variety	color_Yellow	color_Red	color_White	size_Medium	size_Large
1	Donut	1	0	0	0	0
2	Redhaven	0	1	0	0	1
3	Nectarine	0	1	0	1	0
4	White	0	0	1	0	1

	size_Small	taste_Sweet	taste_Tangy	fuzziness_Fuzzy	fuzziness_Smooth
1	1	0	1	1	0
2	0	1	0	1	0
3	0	1	0	0	1
4	0	1	0	1	0

Unstructured Data as Numerical Data: Example

Benefits of peaches

Nutrients that can aid in heart health, gut health and immune function are all found in peaches.

For starters, the fiber in peaches — including both soluble and insoluble fiber — provides health benefits. "Soluble fiber can stabilize blood sugars and keeps cholesterol level in check," Zumpano says, "and then insoluble fiber more aids in digestion and helps prevent constipation."

Most of us in the U.S. get far below the recommended 25 to 40 grams of fiber per day, Derocha adds, and peaches can be a delicious way to get a fiber boost.

Peaches have "a good amount of potassium," Derocha says, which is "a mineral that helps regulate blood pressure and helps with muscle and nerve function." Animal studies have also shown that peach extract may help lower cholesterol and blood pressure, Zumpano says.

Finally, peaches contain a good dose of vitamin C, which can support your immune system, Derocha says. While other fruits (like strawberries and kiwi) contain more, the vitamin C in peaches is a nice bonus in a fruit already packed with other nutrients.

Source: The Today Show, [Peach Benefits](#)

Unstructured Data as Numerical Data: Example

```
# Take the paragraph from the Today Show about the benefits of peaches:
peach_paragraph <- (
  "Nutrients that can aid in heart health, gut health and immune function are all found in peaches.

  For starters, the fiber in peaches - including both soluble and insoluble fiber -
  provides health benefits.
  'Soluble fiber can stabilize blood sugars and keeps cholesterol level in check,'
  Zumpano says, 'and then insoluble fiber more aids in digestion and helps prevent constipation.'

  Most of us in the U.S. get far below the recommended 25 to 40 grams of fiber per day,
  Derocha adds, and peaches can be a delicious way to get a fiber boost.

  Peaches have 'a good amount of potassium,' Derocha says, which is
  'a mineral that helps regulate blood pressure and helps with muscle and nerve function.'
  Animal studies have also shown that peach extract may help lower cholesterol and blood pressure,
  Zumpano says.

  Finally, peaches contain a good dose of vitamin C, which can support your immune system,
  Derocha says.
  While other fruits (like strawberries and kiwi) contain more,
  the vitamin C in peaches is a nice bonus in a fruit already packed with other nutrients."
)
```


Unstructured Data as Numerical Data: Example

```
library(stringr)

# Split the paragraph up into a vector of unique words with counts
# and remove punctuation and paragraph breaks:
peach_paragraph %>%
  # Remove punctuation and paragraphs
  str_remove_all('[:punct:][:digit:]\r\n') %>%
  # Remove stopwords
  str_remove_all('\\b(the|and|a|to|in|of|is|that|can|for|with|as|it|are|this|by|on|from)\\b') %>%
  # Remove any excess white space
  str_squish() %>%
  strsplit(" ") %>%
  unlist() %>%
  tolower() %>%
  table() %>%
  sort(decreasing = TRUE) %>%
  head(20)
```

fiber	peaches	blood	derocha	health	helps
6	4	3	3	3	3
says	c	cholesterol	contain	function	get
3	2	2	2	2	2
good	have	immune	insoluble	more	nutrients
2	2	2	2	2	2
other	pressure				
2	2				

Rectangular ('Tabular') Data

Units and Observations

In rectangular data, rows are observations.

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 - Take a slice (cross-section) at a single point in time
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This is a really important distinction (we'll come back to it later)

Variables or Features

The columns of a rectangular dataset typically indicate 'variables' or 'features.'

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To this point, all rectangular datasets should have a **primary key**: A variable which uniquely identifies each observation.

This could be implicit – a combination of two or more variables – or explicit, like a unique ID.

This is **extremely important** – we will come back to it.

Numerical Variables: Continuous vs. Discrete

We've actually already seen that numerical data, like peaches, can come in many different flavours:

1. **Continuous**: Can take any value within a range:
 - **Interval**: Differences have understood meaning, but no absolute zero (e.g., time picked).
 - **Ratio**: Differences have understood meaning but there is an absolute zero (e.g., weight).

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1. **Continuous:** Can take any value within a range:
 - **Interval:** Differences have understood meaning, but no absolute zero (e.g., time picked).
 - **Ratio:** Differences have understood meaning but there is an absolute zero (e.g., weight).
2. **Discrete:** Can only take specific values, often whole numbers:
 - **Count:** Non-negative integers representing the number of occurrences (e.g., number of peaches).
 - **Ordinal:** A meaningful order but no consistent meaningful difference between values (e.g., rankings).
 - **Nominal:** Categories with no inherent order (e.g., varieties of peach).
 - **Binary:** A special case of nominal data with only two categories (e.g. red peach).

Putting it all Together: 'Tidy' Data

Tidy data is data where:

1. Each observation is a row
2. Each variable is a column
3. Each cell is a value

country	year	cases	population
Afghanistan	1999	1815	19997071
Afghanistan	2000	3666	20005360
Brazil	1999	3737	17206362
Brazil	2000	8488	17404898
China	1999	21258	1272015272
China	2000	21766	128006583

variables

country	year	cases	population
Afghanistan	1999	1815	19997071
Afghanistan	2000	3666	20005360
Brazil	1999	3737	17206362
Brazil	2000	8488	17404898
China	1999	21258	1272015272
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observations

country	year	cases	population
Afghanistan	1999	1815	19997071
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China	1999	21258	1272015272
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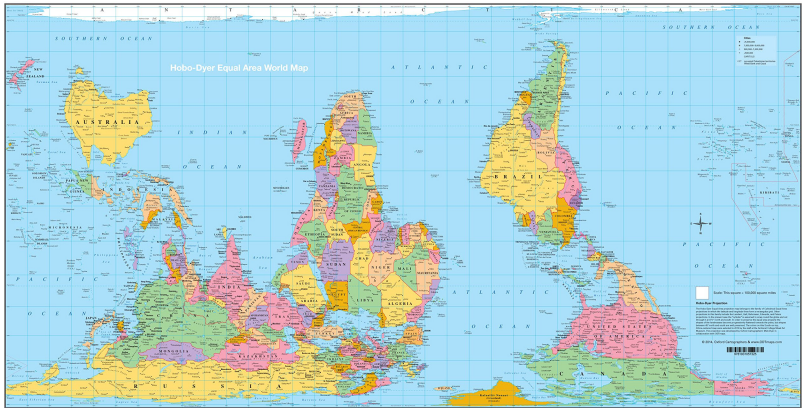
values

Source: Hadley Wickham, [Data Tidying](#)

Example: Geographic Data as Tidy Data



Example: Geographic Data as Tidy Data



Geographic Data as Tidy Data

```
library(sf)

# read in shapefile:
shp <- st_read("../data/ne_10m_admin_0_countries/ne_10m_admin_0_countries.shp")
head(shp)
```

Simple feature collection with 6 features and 168 fields

Geometry type: MULTIPOLYGON

Dimension: XY

Bounding box: xmin: -109.4537 ymin: -55.9185 xmax: 140.9776 ymax: 7.35578

Geodetic CRS: WGS 84

	featurecla	scalerrank	LABELRANK	SOVEREIGNT	SOV_A3	ADM0_DIF	LEVEL		TYPE	TLC	ADMIN	ADM0_A3
1	Admin-0 country	0	2	Indonesia	IDN	0	2	Sovereign country	1	Indonesia	IDN	
2	Admin-0 country	0	3	Malaysia	MYS	0	2	Sovereign country	1	Malaysia	MYS	
3	Admin-0 country	0	2	Chile	CHL	0	2	Sovereign country	1	Chile	CHL	
4	Admin-0 country	0	3	Bolivia	BOL	0	2	Sovereign country	1	Bolivia	BOL	
5	Admin-0 country	0	2	Peru	PER	0	2	Sovereign country	1	Peru	PER	

Example: Geographic Data as Tidy Data

```
library(sf)

# read in shapefile:
shp <- st_read("../data/ne_10m_admin_0_countries/ne_10m_admin_0_countries.shp")
head(shp$geometry)
```

```
Geometry set for 6 features
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -109.4537 ymin: -55.9185 xmax: 140.9776 ymax: 7.35578
Geodetic CRS:  WGS 84
First 5 geometries:
```

```
MULTIPOLYGON (((117.7036 4.163415, 117.7036 4.1...
MULTIPOLYGON (((117.7036 4.163415, 117.6971 4.1...
MULTIPOLYGON (((-69.51009 -17.50659, -69.50611 ...
MULTIPOLYGON (((-69.51009 -17.50659, -69.51009 ...
MULTIPOLYGON (((-69.51009 -17.50659, -69.63832 ...
```


Geographic Data as Tidy Data

```
library(sf)

# read in shapefile:
shp <- st_read("data\\ne_10m_admin_0_countries\\ne_10m_admin_0_countries.shp")
shp$geometry[1][[1]]

MULTIPOLYGON (((117.7036 4.163415, 117.7036 4.163415, 117.7381 4.157242, 117.7836 4.157242, 117.8526 4.157242,
4.156683, 117.9121 4.144924, 117.9182 4.100043, 117.9348 4.059882, 117.9014 4.036689, 117.8871 4.031928, 117.87
4.031562, 117.8386 4.040188, 117.8191 4.068671, 117.7627 4.100735, 117.7394 4.13231, 117.7036 4.163415))), ((124
-9.18019, 124.4658 -9.179376, 124.5178 -9.17783, 124.5632 -9.170831, 124.6043 -9.156345, 124.668 -9.113051, 124
-9.107517, 124.6732 -9.099298, 124.6836 -9.074802, 124.688 -9.069513, 124.6916 -9.067804, 124.7581 -9.047947, 1
-9.042901, 124.7888 -9.018976, 124.825 -9.008071, 124.8403 -9.001072, 124.8699 -8.994399, 124.8971 -8.974542, 1
-8.962091, 124.9195 -8.962016, 124.9225 -8.986324, 124.9092 -9.020327, 124.9079 -9.037483, 124.913 -9.05278, 12
-9.065492, 124.933 -9.074484, 124.9479 -9.078515, 124.9659 -9.077274, 124.9789 -9.072003, 125.0044 -9.05557, 12
-9.035003, 125.0653 -9.023221, 125.0686 -9.015883, 125.073 -8.996452, 125.0765 -8.990768, 125.0868 -8.98622, 12
-8.988598, 125.0953 -8.994799, 125.1025 -9.00131, 125.1213 -9.011852, 125.1394 -9.024771, 125.154 -9.040791, 12
-9.060634, 125.162 -9.082339, 125.145 -9.145177, 125.147 -9.154479, 125.1512 -9.165021, 125.1527 -9.170824, 125
-9.17577, 125.1506 -9.185381, 125.1425 -9.189102, 125.1179 -9.187552, 125.107 -9.188482, 125.0915 -9.19675, 125
-9.205535, 125.0739 -9.211323, 125.0597 -9.210496, 125.0534 -9.206672, 125.0414 -9.193236, 125.0345 -9.188172,
-9.186312, 125.0072 -9.186105, 124.9745 -9.191893, 124.9579 -9.211943, 124.9541 -9.241502, 124.9602 -9.276022,
-9.293385, 124.9728 -9.304651, 124.9967 -9.325838, 125.0179 -9.338964, 125.0234 -9.345578, 125.0248 -9.351986,
-9.367696, 125.0238 -9.374724, 125.0463 -9.409864, 125.0509 -9.424697, 125.0616 -9.485772, 125.0613 -9.486017,
-9.507501, 125.002 -9.53631, 124.9866 -9.577325, 124.9797 -9.649021, 124.9707 -9.662367, 124.8748 -9.724705, 12
```

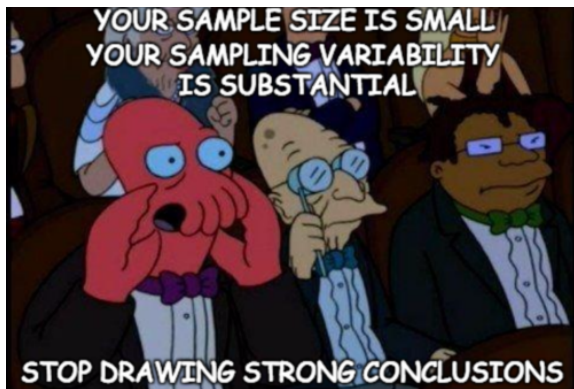
Aside: 'Big' Data

A brief but important aside: You've probably heard of 'big' data.

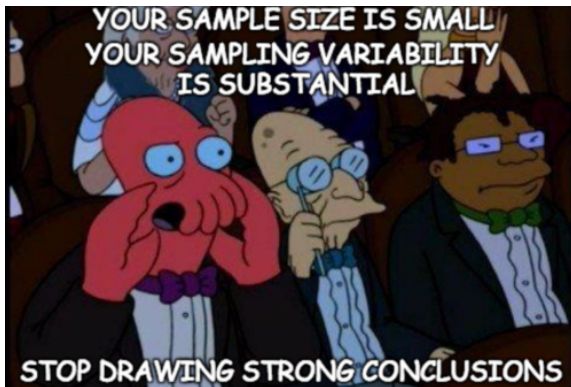
Big data 'as large n ': a very large number of observations

Big data 'as large p ': a very large number of variables/features

Aside: 'Big' Data



Aside: 'Big' Data



Aside: Databases

In most professional settings, you will get data from a **database system**.

Database system: an organized collection of data that is stored and accessed via a computer

- The way a database is organized is called a **schema**
- Since a database is used for data *storage*, a user typically “reads” and “writes” to a database based on permissions
- Read and write via **queries**
- Queries are often constructed/written in **domain-specific languages** like SQL, but not always
- A user can typically read and write via R

Aside: Relational Databases

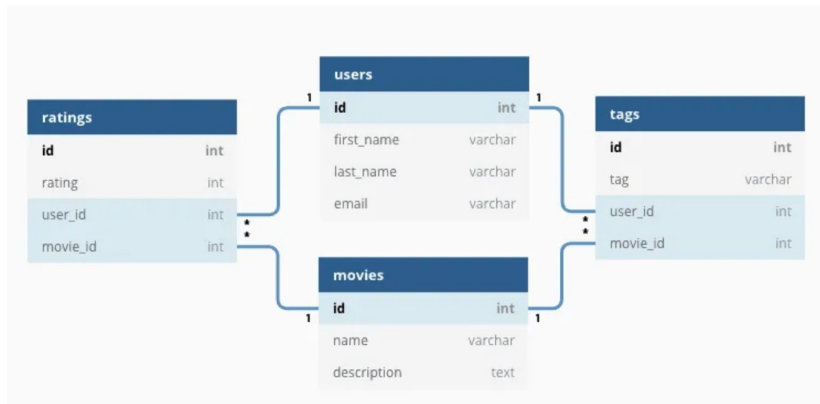
Relational databases

- Data is stored in multiple tables to avoid redundancy
- Tables are linked based on common **keys**
- SQL is used to access data

Non-relational databases

- Data stored in a way that is not based on tabular relations (e.g. MongoDB uses JSON like documents)
- Data is accessed using a wide variety of (sometimes customised) languages

Aside: Relational Databases



Source: Ethan Duong, [Relational Databases](#)

Reading and Storing Data

Data Building Blocks in R and python

In both R and python, data is stored in objects.

However, they differ in a key respect: 'mutability'

- R objects are *immutable*: once created, they cannot be changed in place
- R uses a copy-on-modify logic – if you change an object, a new object is created in memory
- In python some objects are *immutable* while others are *mutable*: they can be changed in place

Data in R and python

The basic data objects types in R and python are similar, but confusingly different:

R	Python	R Example	Python Example	Mutable (Py)?
Numeric	float	<code>x <- 3.14</code>	<code>x = 3.14</code>	No
Integer	int	<code>x <- 42L</code>	<code>x = 42</code>	No
Character	str	<code>x <- "hello"</code>	<code>x = "hello"</code>	No
Logical	bool	<code>x <- TRUE</code>	<code>x = True</code>	No
Vector	list	<code>x <- c(1, 2, 3)</code>	<code>x = [1, 2, 3]</code>	Yes
List	list	<code>x <- list(1, "a")</code>	<code>x = [1, "a"]</code>	Yes
—	tuple	—	<code>x = (1, "a")</code>	No
Named List	dict	<code>list(a=1, b=2)</code>	<code>{"a": 1, "b": 2}</code>	Yes

Remember, in R none of these are mutable! Everything is copy-on-modify.

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Named List	dict	<code>list(a=1, b=2)</code>	<code>{"a": 1, "b": 2}</code>	Yes

Remember, in R none of these are mutable! Everything is copy-on-modify. **Note:** most of these objects are not rectangular data objects! We'll get there...

(Im)mutability in Practice: R

Consider two names bound to the same object:

```
x <- c(1, 2, 3)
y <- x
obj_addr(y) == obj_addr(x)
```

```
[1] TRUE
```

(Im)mutability in Practice: R

Consider two names bound to the same object:

```
x <- c(1, 2, 3)
y <- x
obj_addr(y) == obj_addr(x)
```

```
[1] TRUE
```

Now modify x:

```
x[3] <- 4
y
```

```
[1] 1 2 3
```

(Im)mutability in Practice: R

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```
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obj_addr(y) == obj_addr(x)
```

```
[1] TRUE
```

Now modify x:

```
x[3] <- 4
y
```

```
[1] 1 2 3
```

```
obj_addr(y) == obj_addr(x)
```

```
[1] FALSE
```

(Im)mutability in Practice: python

Consider a string in python, which is immutable:

```
x = "hello"  
y = x  
id(x) == id(y)
```

True

(Im)mutability in Practice: python

Consider a string in python, which is immutable:

```
x = "hello"  
y = x  
id(x) == id(y)
```

True

Modify x:

```
x = "world"  
y
```

'hello'

(Im)mutability in Practice: python

Consider a string in python, which is immutable:

```
x = "hello"  
y = x  
id(x) == id(y)
```

True

Modify x:

```
x = "world"  
y
```

'hello'

```
id(x) == id(y)
```

False

(Im)mutability in Practice: python

Now consider a list in python, which is mutable:

```
x = [1, 2, 3]
y = x
id(x) == id(y)
```

True

(Im)mutability in Practice: python

Now consider a list in python, which is mutable:

```
x = [1, 2, 3]
y = x
id(x) == id(y)
```

True

Modify x:

```
x[2] = 4
y
```

[1, 2, 4]

(Im)mutability in Practice: python

Now consider a list in python, which is mutable:

```
x = [1, 2, 3]
y = x
id(x) == id(y)
```

True

Modify x:

```
x[2] = 4
y
```

```
[1, 2, 4]
```

```
id(x) == id(y)
```

True

Rectangular Data Objects in R (and python)

Eventually we want to work with 'rectangular' data objects:

Concept	R Example	Python Example
Matrix	<code>matrix(1:6, nrow=2)</code>	<code>np.array([[1, 2, 3], [4, 5, 6]])</code>
Array	<code>array(1:8, dim=c(2,2,2))</code>	<code>np.array([[[1,2], [3,4]], [[5,6], [7,8]]])</code>
Data Frame	<code>data.frame(a=1:3, b=4:6)</code>	<code>pd.DataFrame({"a": [1,2,3], "b": [4,5,6]})</code>

File Formats for Rectangular Data

Common file formats for storing tabular data:

- Comma-separated values (.csv) – ubiquitous and simple
 - Each *line* is an observation
 - Each variable value is separated by a comma
- Application specific (proprietary) formats (.dta, .sav, .xlsx etc.)
 - Can allow for richer representations including meta-data
 - More complex, and not necessarily human-readable

Often choice is dictated by the source (and size) of the data

Packages like `haven` in R allow for reading in non-csv formats.

Ingesting Data in R

Recall that we saved `peach_counts.csv` a while back.

In base R we can ingest that `.csv` using the function `read.csv()`:

```
peach_counts <- read.csv("../data/peach_counts.csv",  
                          header = TRUE,  
                          stringsAsFactors = FALSE)  
head(peach_counts)
```

	variety	count	gbp_per_lb
1	Donut	150	2.0
2	Redhaven	200	3.0
3	Nectarine	100	3.5
4	White	120	2.5

We could also use `read_csv()` for a tidy implementation.

Transforming and Manipulating Data

Wrangling Data in Base R

To work with data in base R, we will typically have to manipulate objects directly:

```
# Add a new variable
peach_counts$revenue <- peach_counts$count * peach_counts$gbp_per_lb
# Keep only these columns
peach_counts <- peach_counts[, c("variety", "revenue")]
# Sort by revenue in descending order
peach_counts <- peach_counts[order(-peach_counts$revenue), ]

head(peach_counts)
```

	variety	revenue
2	Redhaven	600
3	Nectarine	350
1	Donut	300
4	White	300

Wrangling Data in Tidy R

Tidy R gives us an alternative approach.

`{dplyr}` gives us useful and literal tools for wrangling data in R:

- `mutate()`: Add or modify variables in a data frame
- `select()`: Choose specific columns from a data frame
- `filter()`: Subset rows based on conditions
- `arrange()`: Sort rows by one or more variables
- and many more (also see other tidyverse packages)

Wrangling Data in Tidy R

Using the pipe `%>%` or `|>` (native pipe) allows us to chain operations:

```
peach_counts <- read.csv("../data/peach_counts.csv",  
                          header = TRUE,  
                          stringsAsFactors = FALSE)  
  
peach_counts %>%  
  # Add a new variable  
  mutate(revenue = count * gbp_per_lb) %>%  
  # Keep only these columns  
  select(variety, revenue) %>%  
  # Sort by revenue in descending order  
  arrange(desc(revenue))
```

	variety	revenue
1	Redhaven	600
2	Nectarine	350
3	Donut	300
4	White	300

Grouping and Hierarchies

Sometimes data has a nested structure.

This can occur in different ways:

1. Repeated observations of the same units:
 - each observation is *nested* under a single unit
2. Hierarchical data:
 - each unit is *nested* under a higher-level unit (cluster)
3. Binned data:
 - each observation is *nested* under a bin based on some variable

We may sometimes want to restructure our data given that hierarchy.

Grouping in R (Tidyverse)

Consider data where each *unit* belongs to some hierarchy:

```
# Example dataset (unique peach = unit)
peach_weights <- data.frame(
  variety = sample(c("Donut", "Redhaven", "Nectarine", "White"),
                  100, replace = TRUE),
  weight = runif(100, 0.5, 1.5)
)

head(peach_weights)
```

	variety	weight
1	White	0.6308958
2	Nectarine	0.7760415
3	Redhaven	1.3246552
4	Redhaven	1.4691023
5	White	1.3579437
6	Redhaven	0.5220077

Grouping in R (Tidyverse)

We can group by a higher variable and summarise across that variable:

```
# Group by 'variety' and summarise:
peach_weights %>%
  group_by(variety) %>%           # variable on which to group
  summarise(                     # summarise across grouping var
    count = n(),                 # counts observations
    total_weight = sum(weight),  # apply a function (sum(x))
    mean_weight = mean(weight)  # a different function (mean(x))
  )
```

```
# A tibble: 4 x 4
  variety    count total_weight mean_weight
  <chr>      <int>      <dbl>      <dbl>
1 Donut         30         28.3         0.943
2 Nectarine     17         17.5         1.03
3 Redhaven      24         25.4         1.06
4 White         29         29.2         1.01
```

Reshaping in R

Now consider data with multiple *observations* per *unit* (e.g., yield per peach variety over time):

```
# Example dataset (peach variety = unit, observed over 25 time periods)
peach_panel <- data.frame(
  variety = rep(c("Donut", "Redhaven", "Nectarine", "White"), 25),
  year = rep(2000:2024, each = 4),
  yield = runif(100, 50, 200)
)

head(peach_panel)
```

	variety	year	yield
1	Donut	2000	159.40708
2	Redhaven	2000	84.10409
3	Nectarine	2000	137.95510
4	White	2000	152.53014
5	Donut	2001	199.15108
6	Redhaven	2001	82.85935

Reshaping in R

Let's reshape our data from long to wide using {tidyr}:

```
# Reshape from long to wide:
peach_panel_wide <- peach_panel %>%
  pivot_wider(
    names_from = year,          # variable to use as column names
    values_from = yield,        # variable to use as values
    values_fill = list(yield = NA) # fill missing values with NA
  )

head(peach_panel_wide)
```

```
# A tibble: 4 x 26
```

	variety	`2000`	`2001`	`2002`	`2003`	`2004`	`2005`	`2006`	`2007`	`2008`	`2009`
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Donut	159.	199.	193.	188.	74.0	170.	75.9	104.	55.4	115
2	Redhaven	84.1	82.9	96.0	182.	82.5	128.	94.9	165.	130.	137
3	Nectarine	138.	87.1	86.4	69.9	110.	88.4	134.	148.	158.	147
4	White	153.	195.	86.6	175.	83.3	133.	77.1	170.	73.5	194

```
# i 15 more variables: `2010` <dbl>, `2011` <dbl>, `2012` <dbl>, `2013` <dbl>,
#   `2014` <dbl>, `2015` <dbl>, `2016` <dbl>, `2017` <dbl>, `2018` <dbl>,
#   `2019` <dbl>, `2020` <dbl>, `2021` <dbl>, `2022` <dbl>, `2023` <dbl>,
#   `2024` <dbl>
```


Reshaping in R

There and back again (back to long format):

```
# Reshape from wide to long:
peach_panel_wide %>%
  pivot_longer(
    cols = -variety,           # keep 'variety' as is
    names_to = "year",        # new column for years
    values_to = "yield"       # new column for yield values
  ) %>%
  head()
```

```
# A tibble: 6 x 3
  variety year  yield
  <chr>   <chr> <dbl>
1 Donut  2000   159.
2 Donut  2001   199.
3 Donut  2002   193.
4 Donut  2003   188.
5 Donut  2004    74.0
6 Donut  2005   170.
```

Merges and Joins in R

Recall our relational database setting – we had multiple tables connected by keys.

In R, we can merge (join) a left object and a right object together using the same principles:

- **Inner join**: Keep only rows with matching keys in both tables
- **Left join**: Keep all rows from the left table, and matching rows from the right table
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- **Full join**: Keep all rows from both tables

Merges and Joins in R

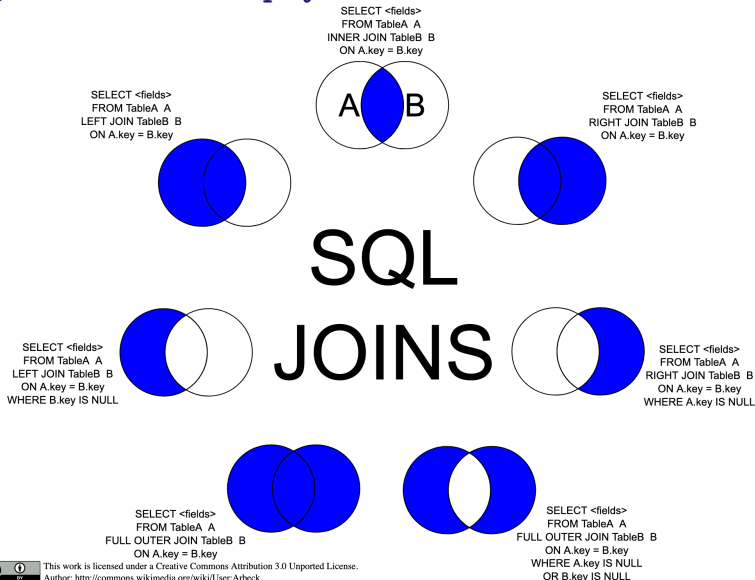
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It is very wise to **check** your data after merges. Bad merges can create **very bad** outcomes.

Merges and Joins: {dplyr} and SQL



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Author: <http://commons.wikimedia.org/wiki/User:Arbeck>

Merges and Joins in R

Recall our earlier `peach_features` object? Let's join it to our `peach_counts` object.

Our **key** is the `variety` variable, present in both objects:

```
# Join peach_counts to peach_features:
peach_joined <- peach_counts %>%
  # Join on 'variety'
  left_join(peach_features, by = "variety")
```

```
peach_joined
```

	variety	count	gbp_per_lb	color	size	taste	fuzziness
1	Donut	150	2.0	Yellow	Small	Tangy	Fuzzy
2	Redhaven	200	3.0	Red	Large	Sweet	Fuzzy
3	Nectarine	100	3.5	Red	Medium	Sweet	Smooth
4	White	120	2.5	White	Large	Sweet	Fuzzy

Visualising Data

What is Visualisation?

Visualisation: Presenting something visually (wow, insightful).

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This could mean a:

- histogram or density plot of a variable
- two-way (or three-way) scatterplot of points
- map of something
- photo or screenshot of an experimental intervention
- coefficient plot
- trend-line over time
- table (!)

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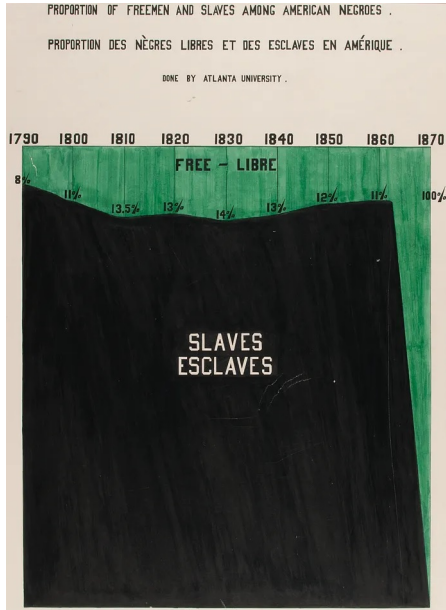
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Visualisation is important through the life-cycle of a project:

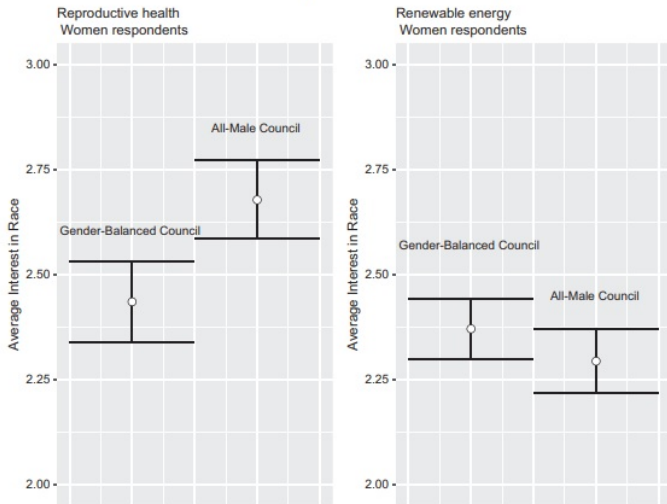
- **Early:** looking at data to assess quality, properties, face validity
- **Mid:** monitoring what you are doing along the way
- **Late:** what you include in presentations/papers

Good Visualisation, Bad Visualisation



Good Visualisation, Bad Visualisation

FIGURE 2. Treatment Effects for Women Respondents on Interest in Hypothetical Race



Note: Error Bars at 95% confidence intervals. See also Model 1 of [Table 1](#).

Good Visualisation, Bad Visualisation

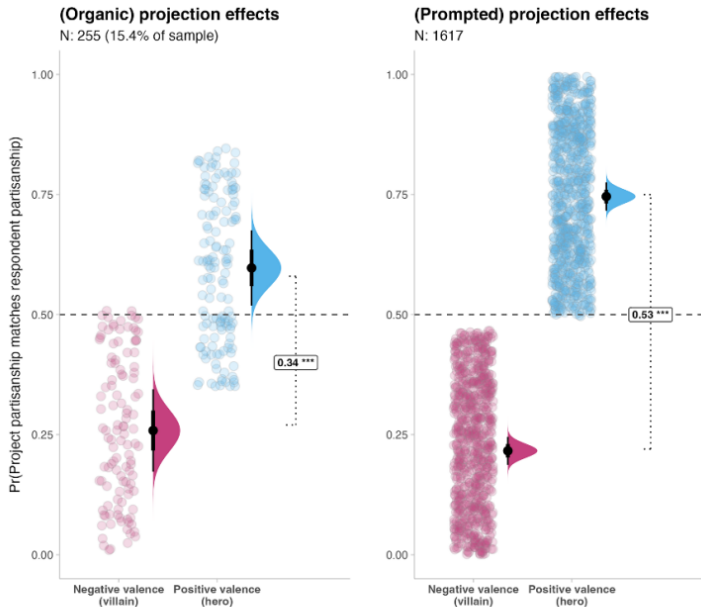
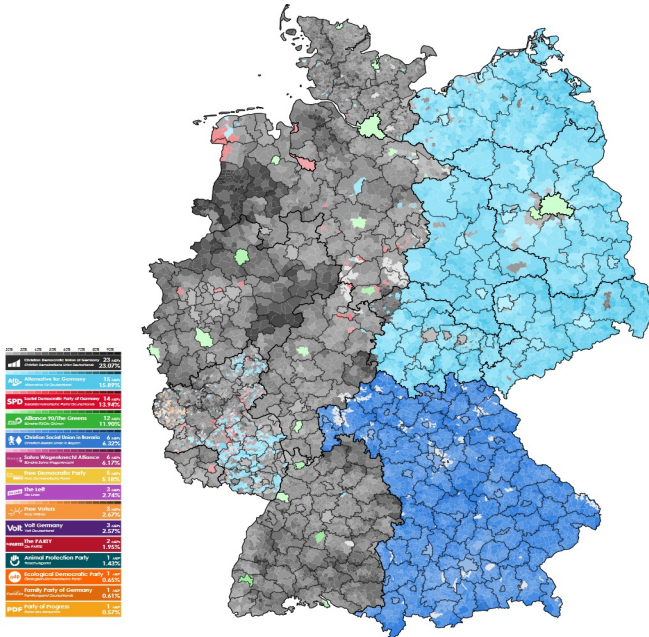
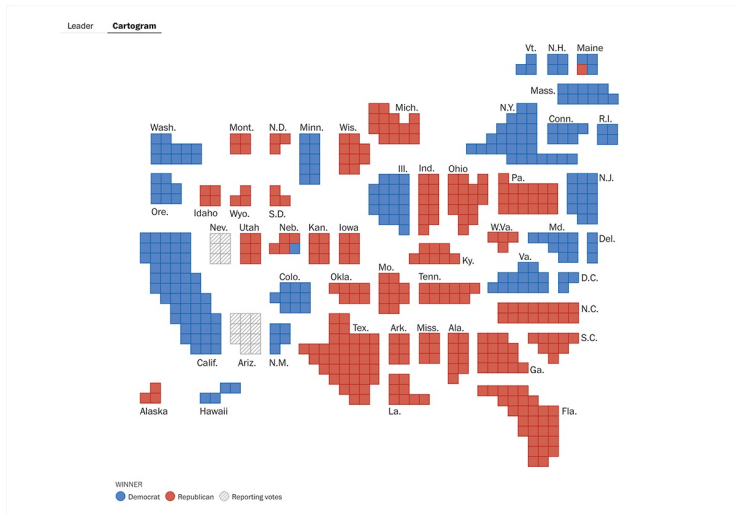


Figure 5. Modeling projecting via false recall (Study 2).

Good Visualisation, Bad Visualisation



Good Visualisation, Bad Visualisation



The Language of Visualisation: Some Anatomy

A visualisation is some visual representation drawn on a plane, with four necessary components.

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 - Shape

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A visualisation is a **mapping** of **data** to **aesthetics** via visual **geometries** to produce meaning. Formally this can be represented as a “**grammar of graphics**” (Wilkinson, 2005).

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Mappings of data to aesthetics and geometries give visualisations **meaning** (or 'content'). Note what, strictly speaking, doesn't matter. . .

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Of course this stuff **does matter**, but not in the way we imagine. Let's return to it later.

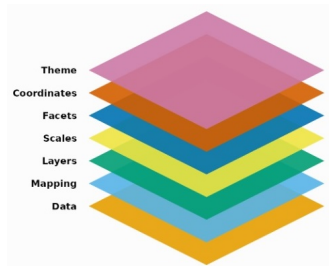
Grammar of Graphics $\longleftrightarrow$ {ggplot2}

In {ggplot2}, three key pieces:

1. **Data**: The information
2. **Mapping**: Correspondence between data and aesthetics
3. **Layers**: Geometry and transformation

Then we get:

- Scales: Paired to an aesthetic
- Facets: Subsets of data
- Coordinates: The x-y system of the canvas
- Theme: Design choices



Source: [Introduction to ggplot2](#)

The Good, The Bad, and the Ugly

Misconception: Good data visualisation = good visual taste.

The Good, The Bad, and the Ugly

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No! Good (and bad) data visualisation is **not** primarily about how nice (ugly) your plot looks.

It is about **communication** with the **reader/viewer**.

Making your plot look nice is the **last step** in data/model visualisation, and only relevant once your visualisation meets its “communication goal.”

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Making your plot look nice is the **last step** in data/model visualisation, and only relevant once your visualisation meets its “communication goal.”

If your visualisation is misleading, confusing, or fails to communicate, then from a scientific perspective it **does not matter** how pretty it is.

Design and explanation only matter inasmuch as they contribute to the communication goal.

Goals of Visualisation

What are useful **communication goals**?

- Delivering a (set of) fact(s)
- Telling a story
- Imparting meaning

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- Delivering a (set of) fact(s)
- Telling a story
- Imparting meaning

So, what makes for a good (bad) visualisation:

- **Thoughtful**: There is a clear communication goal, and the visualisation is built with that goal in mind.
- **Honest**: The reader is not misled about the data or the meaning.

Good vs. Bad: General Principles

Emphasis: **thoughtful** and **honest** visualisation.

Thoughtful:

- Communication goal you can articulate in words
- One aesthetic mapping per variable (some exceptions apply)
- No unmapped aesthetics
- Visually interpretable mappings
- Balance information and meaning – raw information is rarely meaningful
- Avoid clutter (“chartjunk” in Tufte’s (1983) words)

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Honest: - Avoid distortions of the data or statistics - Avoid axis compression where possible/necessary - Avoid “made up” data

Good vs. Bad: The Lie Factor

Tufte (1983) proposes the Lie Factor:

$$\text{Lie Factor} = \frac{\text{Size of Comparison in Visual}}{\text{Size of Comparison in Data}}$$

Where:

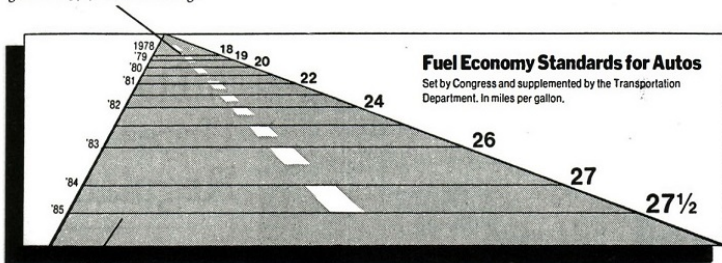
$$\text{Size of Comparison} = \frac{\text{Second Value} - \text{First Value}}{\text{First Value}}$$

Essentially, any Lie Factor other than 1 implies a visual distortion of the data.

Good vs. Bad: The Lie Factor

Classic case from the New York Times:

This line, representing 18 miles per gallon in 1978, is 0.6 inches long.



This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

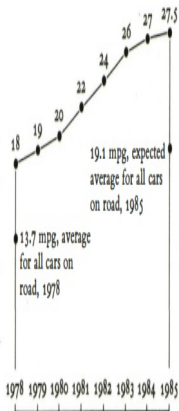
Tufte calculates the Lie Factor of this as $14.8 = \frac{783\%}{53\%}$

Good vs. Bad: The Lie Factor

Undoing the lie, mostly (Tufte, 1983):

REQUIRED FUEL ECONOMY STANDARDS:

NEW CARS BUILT FROM 1978 TO 1985



Visualising Our Peaches

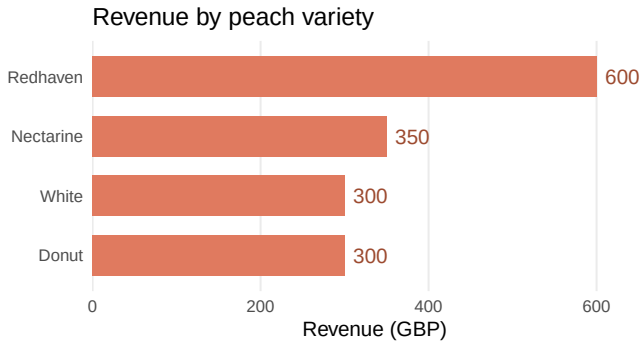
We have now built up several peachy datasets. Let's do some visualisations with `{ggplot2}`.

Visualising Our Peaches: Ranking Revenue

Which variety earns the most? Here we want to simply map revenue to bar length, and then order the bars from highest revenue to lowest:

```
revenue_plot <- peach_counts %>%  
  # Revenue combines count and price  
  mutate(revenue = count * gbp_per_lb) %>%  
  # Aesthetics: revenue as bar length, varieties ordered by revenue  
  ggplot(aes(x = revenue, y = reorder(variety, revenue))) +  
  geom_col(fill = "#E07A5F", width = 0.68) +  
  # Label each bar directly, so no legend is needed  
  geom_text(aes(label = revenue), hjust = -0.25, colour = "#9C4A2F") +  
  scale_x_continuous(expand = expansion(mult = c(0, 0.12))) +  
  labs(title = "Revenue by peach variety",  
        x = "Revenue (GBP)", y = NULL) +  
  theme_minimal(base_size = 16) +  
  theme(panel.grid.major.y = element_blank(),  
        panel.grid.minor = element_blank())
```

Visualising Our Peaches: Ranking Revenue

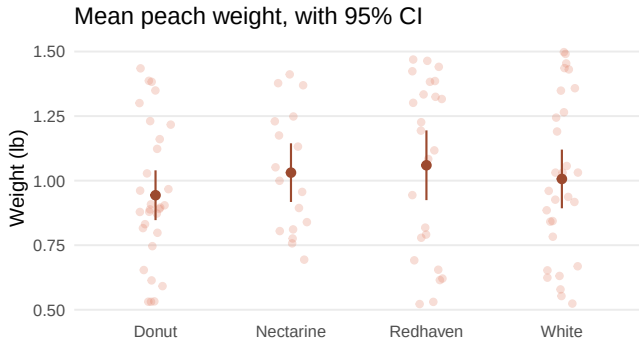


Visualising Our Peaches: The Average Weight

Which peaches are heaviest? Each of our 100 peaches has a weight, so we plot those and average by variety, adding a confidence interval on top:

```
weight_plot <- peach_weights %>%  
  # Aesthetics: variety on x, weight on y  
  ggplot(aes(x = variety, y = weight)) +  
  # Faded raw points: one peach each  
  geom_jitter(width = 0.12, height = 0, alpha = 0.25,  
              colour = "#E07A5F") +  
  # Mean and 95% CI (mean +/- 1.96 SE) per variety  
  stat_summary(fun.data = mean_se, fun.args = list(mult = 1.96),  
              geom = "pointrange", colour = "#9C4A2F",  
              linewidth = 0.8, size = 0.5) +  
  labs(title = "Mean peach weight, with 95% CI",  
        x = NULL, y = "Weight (lb)") +  
  theme_minimal(base_size = 16) +  
  theme(panel.grid.major.x = element_blank(),  
        panel.grid.minor = element_blank())
```

Visualising Our Peaches: The Average Weight

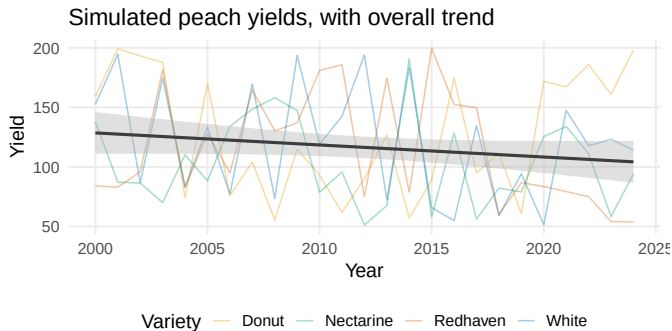


Visualising Our Peaches: Yields Over Time

How does peach yield trend over time? We can plot the individual time-series (yields per year, per variety), and then overlay a linear time trend (a linear regression):

```
yield_plot <- peach_panel %>%  
  # Aesthetics: year on x, yield on y  
  ggplot(aes(x = year, y = yield)) +  
  # A faded line for each variety  
  geom_line(aes(colour = variety, group = variety),  
            alpha = 0.35, linewidth = 0.6) +  
  scale_colour_manual(values = c(  
    Donut = "#E69F00", Redhaven = "#D55E00",  
    Nectarine = "#009E73", White = "#0072B2")) +  
  # One bold overall trend, pooled across all varieties  
  geom_smooth(method = "lm", colour = "#3D3D3D",  
              fill = "grey70", linewidth = 1.2) +  
  labs(title = "Simulated peach yields, with overall trend",  
        x = "Year", y = "Yield", colour = "Variety") +  
  theme_minimal(base_size = 16) +  
  theme(panel.grid.minor = element_blank(),  
        legend.position = "bottom")
```

Visualising Our Peaches: Yields Over Time



Summary and Next Steps

Today we covered:

1. What is data?
2. Different kinds of data
3. Working with data in R (and python)
4. Visualisation of data in R

This afternoon you will practice 2 and 3.

Tomorrow:

1. Probability
2. Data generating processes